

# CHRONUS: A MEMORY-TRAINED AGENTIC AI FOR HUMAN PERSONALITY AND MINDSET SIMULATION

A Project Report

Submitted By

**Tirth Sakariya**

2303031050723

**Manav Aghera**

2403031057168

**Manish Prajapati**

2303031050721

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Under the Guidance of

**Prof. Shubham Upadhyay**  
Assistant Professor



**Parul<sup>®</sup> University**  
**NAAC A++**

VADODARA

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**PARUL UNIVERSITY**

**CERTIFICATE**

This is to Certify that the Project entitled “**CHRONUS: A Memory-Trained Agentic AI for Human Personality and Mindset Simulation**” has been successfully completed by :

- **TIRTH SAKARIYA – 2303031050723**
- **MANAV AGHERA – 2403031057168**
- **MANISH PRAJAPATI – 2303031050721**

under my guidance in partial fulfillment of the Bachelor of Technology (B.Tech) in Computer Science & Engineering of Parul University in Academic Year 2025–2026.

Date of Submission : \_\_\_\_\_

---

**Prof. Shubham Upadhyay**

**Project Guide & Mentor**

**Dept. of CSE, PIET**

---

**Prof. Namrata Patel**

**& Prof. Vennela Tandra**

**Project Coordinators**

**Dept. of CSE, PIET**

---

**Dr. Shailendra K. Mishra**

**Head of Department**

**Dept. of CSE, PIET**

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*“The single greatest cause of happiness is gratitude.”*

–Auliq-Ice

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**Tirth Sakariya**  
**Manav Aghera**  
**Manish Prajapati**  
**CSE, PIET**  
**Parul University,**  
**Vadodara**

## Abstract

The loss of human knowledge, personality, and wisdom at the end of life represents one of the most profound and irreversible information losses in human history. Despite the existence of vast personal archives in the form of diaries, letters, emails, voice recordings, and digital communications, no system currently exists that can reconstruct and re-animate a person's complete cognitive identity in an interactive, conversational, and voice-authentic manner. This project presents

CHRONUS, a memory-trained agentic AI system designed to simulate the knowledge, mindset, beliefs, communication style, and voice of any individual by training directly on their personal data.

CHRONUS employs a multi-layered technical architecture: personal documents are parsed and sentence-segmented using spaCy, then encoded into 384-dimensional semantic vector embeddings via the Sentence-BERT all-MiniLM-L6-v2 transformer model, and stored in a ChromaDB local vector database. At inference time, user queries are embedded and matched against stored memories using cosine similarity retrieval. Responses are generated using a novel Mix Method that combines an authentic retrieved passage with a persona-flavoured modern explanation and a closing signature, all delivered in the person's cloned voice using ElevenLabs or XTTS-v2 neural text-to-speech synthesis. Additionally, CHRONUS includes eight pre-trained famous personality models — Einstein, Gandhi, Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, and Shakespeare — built from their verified real writings. A structured 25-question interview protocol enables family members and friends to contribute memories when direct personal data is limited.

The system emphasizes factual grounding and local-first privacy. Retrieval-centric response generation minimizes hallucination compared with unconstrained role-play prompting. Results indicate that semantic vector-based retrieval significantly outperforms keyword matching for personal memory Q&A, the Mix Method response style produces the most immersive and believable interaction, and voice cloning reduces the perceived AI-ness of responses to near-human levels. All processing is conducted locally, ensuring complete privacy with no external API calls for core functionality. Future work includes real-time voice streaming, facial avatar integration, RAG-based LLM augmentation, and mobile application development for collaborative family contributions.

**Keywords:** Agentic AI, Memory Simulation, Voice Cloning, Sentence-BERT, ChromaDB, Vector Embeddings, Human Knowledge Preservation, Neural Text-to-Speech, Persona Modelling, Digital Legacy

# Table of Contents

|                                                        |            |
|--------------------------------------------------------|------------|
| <b>Acknowledgements</b>                                | <b>iv</b>  |
| <b>Abstract</b>                                        | <b>v</b>   |
| <b>List of Tables</b>                                  | <b>xi</b>  |
| <b>List of Figures</b>                                 | <b>xii</b> |
| <b>1 Introduction</b>                                  | <b>1</b>   |
| 1.1 Introduction .....                                 | 1          |
| 1.2 Scope of the Project.....                          | 2          |
| 1.3 Aim of the Project.....                            | 2          |
| 1.4 Objectives of the Project.....                     | 3          |
| 1.5 Problem Definition .....                           | 3          |
| 1.5.1 Irreversible Cognitive Identity Loss.....        | 3          |
| 1.5.2 The Voice Gap in AI Systems .....                | 4          |
| 1.5.3 Fragmented Personal Archives .....               | 4          |
| 1.5.4 Hallucination in LLM-Based Persona Systems ..... | 4          |
| 1.6 Organisation of the Report.....                    | 4          |
| <b>2 Literature Review</b>                             | <b>5</b>   |
| 2.1 Introduction to the Literature Review.....         | 5          |
| 2.2 Conversational AI and Digital Legacy Systems.....  | 5          |

|          |                                                                         |           |
|----------|-------------------------------------------------------------------------|-----------|
| 2.2.1    | Replika (Kuyda, 2017).....                                              | 5         |
| 2.2.2    | HereAfter AI (2021) .....                                               | 6         |
| 2.2.3    | StoryFile (2020).....                                                   | 6         |
| 2.2.4    | Soul Machines (2021).....                                               | 6         |
| 2.3      | Semantic Sentence Embedding and Transformer Models .....                | 7         |
| 2.3.1    | BERT (Devlin et al., 2018) .....                                        | 7         |
| 2.3.2    | Sentence-BERT (Reimers and Gurevych, 2019).....                         | 7         |
| 2.3.3    | InstructGPT and Instruction-Tuned Embeddings (Ouyang et al., 2022)..... | 7         |
| 2.4      | Vector Databases and Similarity Search .....                            | 8         |
| 2.4.1    | Approximate Nearest Neighbour Search (Johnson et al., 2017) .....       | 8         |
| 2.4.2    | ChromaDB (Chroma, 2023).....                                            | 8         |
| 2.4.3    | Pinecone and Weaviate (2021–2023).....                                  | 8         |
| 2.5      | Neural Text-to-Speech and Voice Cloning.....                            | 9         |
| 2.5.1    | WaveNet (Oord et al., 2016) .....                                       | 9         |
| 2.5.2    | Tacotron 2 (Shen et al., 2018) .....                                    | 9         |
| 2.5.3    | ElevenLabs Neural Voice Cloning (2023).....                             | 9         |
| 2.5.4    | XTTS-v2 (Casanova et al., 2024).....                                    | 9         |
| 2.6      | Persona Modelling and Anti-Hallucination .....                          | 10        |
| 2.6.1    | GPT-4 and Persona Prompting (OpenAI, 2023) .....                        | 10        |
| 2.6.2    | Retrieval-Augmented Generation (Lewis et al., 2020).....                | 10        |
| 2.6.3    | MemGPT (Packer et al., 2023) .....                                      | 10        |
| 2.7      | Summary of Literature Review and Research Gap .....                     | 11        |
| <b>3</b> | <b>Problem Statement and Research Questions</b>                         | <b>12</b> |
| 3.1      | Problem Statement.....                                                  | 12        |
| 3.2      | Research Questions.....                                                 | 12        |
| 3.3      | Hypotheses .....                                                        | 13        |

|          |                                               |           |
|----------|-----------------------------------------------|-----------|
| 3.4      | Operational Definitions .....                 | 13        |
| 3.5      | Functional Requirements .....                 | 13        |
| 3.6      | Non-Functional Requirements .....             | 14        |
| <b>4</b> | <b>System Design</b>                          | <b>15</b> |
| 4.1      | System Architecture Overview .....            | 15        |
| 4.2      | Frontend Layer .....                          | 16        |
| 4.3      | API Proxy Layer .....                         | 16        |
| 4.4      | Backend Service Layer .....                   | 16        |
| 4.5      | Memory and Retrieval Layer .....              | 16        |
| 4.6      | Database Schema Design .....                  | 17        |
| 4.7      | System Workflow .....                         | 17        |
| 4.8      | Data Flow Diagrams (DFD) .....                | 19        |
| 4.8.1    | DFD Level 0 (Context Diagram) .....           | 19        |
| 4.8.2    | DFD Level 1 .....                             | 19        |
| 4.8.3    | DFD Level 2 .....                             | 19        |
| 4.9      | Conclusion .....                              | 19        |
| <b>5</b> | <b>Methodology</b>                            | <b>22</b> |
| 5.1      | Research and Development Approach .....       | 22        |
| 5.2      | Data Ingestion and File Handling Method ..... | 22        |
| 5.3      | Sentence Segmentation Strategy .....          | 23        |
| 5.4      | Vector Embedding Method .....                 | 23        |
| 5.5      | ChromaDB Storage and Indexing .....           | 23        |
| 5.6      | Interview Augmentation Protocol .....         | 24        |
| 5.7      | Response Generation — The Mix Method .....    | 24        |
| 5.8      | Famous Personality Models .....               | 25        |

|          |                                           |           |
|----------|-------------------------------------------|-----------|
| 5.9      | Security and Privacy Architecture .....   | 25        |
| 5.10     | Testing and Validation Method .....       | 26        |
| 5.11     | Methodology Outcome .....                 | 26        |
| <b>6</b> | <b>Implementation</b>                     | <b>27</b> |
| 6.1      | System Implementation Overview.....       | 27        |
| 6.2      | Technology Stack Used.....                | 27        |
| 6.3      | Layered Architecture Implementation ..... | 28        |
| 6.3.1    | Presentation Layer .....                  | 28        |
| 6.3.2    | Application Layer .....                   | 28        |
| 6.3.3    | Data Layer .....                          | 28        |
| 6.4      | Data Ingestion Implementation.....        | 28        |
| 6.5      | Memory Builder Implementation .....       | 29        |
| 6.6      | Response Generation Implementation .....  | 29        |
| 6.7      | Interview Module Implementation.....      | 30        |
| 6.8      | User Interface Design .....               | 30        |
| 6.8.1    | Design Principles .....                   | 30        |
| 6.8.2    | Screens and Modules .....                 | 30        |
| 6.9      | Famous Personality Implementation.....    | 31        |
| 6.10     | Voice Synthesis Implementation .....      | 31        |
| 6.11     | Implementation Outcome .....              | 32        |
| <b>7</b> | <b>Testing and Results</b>                | <b>36</b> |
| 7.1      | Testing Strategy .....                    | 36        |
| 7.2      | Functional Testing .....                  | 36        |
| 7.2.1    | Functional Testing Summary .....          | 36        |
| 7.3      | Retrieval Accuracy Testing.....           | 36        |

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| 7.4      | Response Quality Evaluation.....                          | 37        |
| 7.5      | Parser Module Testing.....                                | 37        |
| 7.6      | Interview Augmentation Impact .....                       | 38        |
| 7.7      | API Endpoint Testing .....                                | 38        |
| 7.8      | Security and Privacy Validation.....                      | 38        |
| 7.9      | Integration Testing.....                                  | 39        |
| 7.10     | Result Interpretation .....                               | 40        |
| 7.11     | Conclusion.....                                           | 40        |
| <b>8</b> | <b>Future Work</b>                                        | <b>42</b> |
| 8.1      | Overview .....                                            | 42        |
| 8.2      | Real-Time Voice Streaming and Telephony Integration ..... | 42        |
| 8.3      | Facial Avatar Generation .....                            | 42        |
| 8.4      | LLM Integration via Retrieval-Augmented Generation.....   | 43        |
| 8.5      | Emotion Modelling in Voice Synthesis.....                 | 43        |
| 8.6      | Multi-Language Support.....                               | 43        |
| 8.7      | Mobile Application Development .....                      | 43        |
| 8.8      | Advanced Prioritization and Confidence Scoring .....      | 43        |
| 8.9      | Clinically Informed Grief-Safety Protocol .....           | 44        |
| 8.10     | Final Direction.....                                      | 44        |
|          | <b>References</b>                                         | <b>45</b> |

# List of Tables

|     |                                                                 |    |
|-----|-----------------------------------------------------------------|----|
| 2.1 | Feature comparison of existing systems with CHRONUS .....       | 11 |
| 3.1 | Operational Definitions .....                                   | 13 |
| 4.1 | CHRONUS API Endpoints.....                                      | 17 |
| 4.2 | ChromaDB Collection Schema.....                                 | 18 |
| 4.3 | Model History JSON Schema.....                                  | 18 |
| 5.1 | Interview Augmentation Dimensions .....                         | 24 |
| 6.1 | Complete Technology Stack of the CHRONUS System.....            | 27 |
| 7.1 | Functional Testing Summary .....                                | 37 |
| 7.2 | Retrieval Accuracy Results: Semantic vs. Keyword Matching ..... | 37 |
| 7.3 | Response Quality Rubric Results .....                           | 38 |
| 7.4 | Parser Module Testing Results.....                              | 38 |
| 7.5 | Impact of Interview Augmentation on Model Quality .....         | 38 |
| 7.6 | API Endpoint Testing Results .....                              | 39 |

# List of Figures

|     |                                                     |    |
|-----|-----------------------------------------------------|----|
| 4.1 | System Architecture of the CHRONUS Platform.....    | 15 |
| 4.2 | DFD Level 0: Context Diagram .....                  | 19 |
| 4.3 | DFD Level 1: Major System Processes .....           | 20 |
| 4.4 | DFD Level 2: Retrieval and Generation Details ..... | 21 |
| 6.1 | Home Screen and Primary Workflows .....             | 31 |
| 6.2 | Data Ingestion and Upload Interface .....           | 32 |
| 6.3 | Model Training Progress Indicator .....             | 33 |
| 6.4 | Structured Interview Augmentation Interface.....    | 33 |
| 6.5 | Memory-Grounded Conversational Chat Interface.....  | 34 |
| 6.6 | Previously Trained Models History .....             | 34 |
| 6.7 | Famous Personalities Selection Screen.....          | 35 |

# Chapter 1

## Introduction

### 1.1 Introduction

Human civilisation has always been defined by the transmission of knowledge across generations. From the oral traditions of ancient cultures to the written manuscripts of the Renaissance, and from the printing press to the digital age, humanity has continually sought ways to preserve and disseminate the wisdom, experience, and personality of individuals. Yet despite all technological advances, the death of a person still results in the permanent loss of their cognitive identity — the unique combination of memories, beliefs, communication patterns, values, and voice that defined who they were.

The digital era has left behind an unprecedented volume of personal data: diaries, emails, text messages, voice recordings, social media posts, and handwritten letters that have been digitised. These archives represent an extraordinary record of individual human experience. Yet they remain largely static — searchable by keyword, but not truly queryable in the way one would ask a question to a living person. No system currently exists that can read across these materials, understand their semantic content, and respond in the voice and manner of the person they document.

CHRONUS is an agentic AI system that addresses this gap directly. By training on personal data and applying advanced natural language processing, semantic memory storage, and neural voice cloning, CHRONUS creates an interactive AI model that simulates the complete cognitive identity of a human being. The system can answer questions as that person would, using their actual documented thoughts and beliefs, and can deliver its responses in their cloned voice — making the interaction indistinguishable from a real conversation.

This research positions CHRONUS at the intersection of AI systems engineering and grief technology. Unlike purely generative companions, the design objective is not unrestricted creativity;

it is grounded memory reconstruction with explicit provenance behavior. The practical hypothesis is that semantically retrieved memories generate more trustworthy and emotionally stable interactions than free-form role-play prompts.

## 1.2 Scope of the Project

The scope of this project encompasses the full design, development, and evaluation of the CHRONUS system. This includes:

- Development of a multi-format document ingestion pipeline supporting .txt, .md, .pdf, .docx, .csv, .eml, and .json file types.
- Implementation of a semantic memory system using Sentence-BERT vector embeddings stored in ChromaDB.
- Design of an interview augmentation protocol comprising 25 structured questions across six personality dimensions.
- Construction of eight pre-trained famous personality models based on verified historical writings.
- Integration of neural voice cloning using ElevenLabs or XTTS-v2 for authentic voice replication.
- Development of a React-based frontend and Python Flask backend forming a complete, locally-hosted application.
- Evaluation of retrieval accuracy, response quality, and perceived human-likeness of the voice output.

The project does not include the development of real-time video avatar synthesis or fine-tuning of large language models in this phase, though these are identified as future work directions.

## 1.3 Aim of the Project

The primary aim of this project is to design and develop CHRONUS — a memory-trained agentic AI system that accurately simulates the knowledge, mindset, communication style, and voice of any human individual using their personal written records and audio data. The aim encompasses both the technical implementation of the AI pipeline and the ethical design principles that govern its responsible use.

## 1.4 Objectives of the Project

The specific objectives of this project are as follows:

1. To build a universal data ingestion pipeline capable of processing all major personal archive file formats.
2. To implement semantic sentence embedding using Sentence-BERT and cosine similarity retrieval via ChromaDB for memory-grounded response generation.
3. To design and validate a Mix Method response generation framework that combines authentic retrieved passages with persona-flavoured modern explanations.
4. To integrate neural voice cloning so that all AI responses are delivered in the cloned voice of the target individual.
5. To develop eight pre-trained famous personality models (Einstein, Gandhi, Tesla, Marcus Aurelius, Steve Jobs, Marie Curie, Abraham Lincoln, Shakespeare) based exclusively on their verified real writings and documented philosophy.
6. To design a 25-question interview augmentation system enabling family and friends to contribute structured memories when direct personal data is insufficient.
7. To implement a complete locally-hosted web application with React frontend, Node.js proxy, and Python Flask backend.
8. To evaluate the system against anti-hallucination, response relevance, and human-likeness benchmarks.
9. To ensure all processing is conducted locally with no mandatory external API calls, ensuring complete data privacy.

## 1.5 Problem Definition

The core problem addressed by CHRONUS is the permanent, irreversible loss of human knowledge and personality at the time of death. The following specific problem dimensions are identified:

### 1.5.1 Irreversible Cognitive Identity Loss

An estimated 108 billion humans have ever lived on Earth. Each one possessed a unique combination of knowledge, beliefs, communication patterns, and personality — yet none of this information is

AI-accessible in an interactive, conversational form. While written and audio records exist for many individuals, no system translates these records into a living, queryable AI model of the person.

### **1.5.2 The Voice Gap in AI Systems**

Existing AI systems that simulate personalities — such as Replika, HereAfter AI, and GPT-4 persona prompting — respond exclusively through text. The absence of the person’s real voice is a critical limitation: voice carries emotional texture, familiarity, and identity cues that text fundamentally cannot replicate. A bereaved family member cannot feel genuine connection by reading text that imitates their loved one; they need to hear them.

### **1.5.3 Fragmented Personal Archives**

Personal archives exist in dozens of incompatible formats — handwritten diaries scanned to PDF, email archives in .eml format, WhatsApp conversation exports in .txt, Word documents, and voice messages as audio files. No unified system currently exists to ingest all of these, extract their semantic content, and build a coherent memory model from them.

### **1.5.4 Hallucination in LLM-Based Persona Systems**

Current large language models, when prompted to role-play as historical figures, frequently hallucinate — generating plausible-sounding but factually incorrect statements. For a system designed to represent a real human being, hallucination is not merely an inaccuracy; it is a fundamental failure of integrity. CHRONUS addresses this by grounding every response in retrieved, verified source text.

## **1.6 Organisation of the Report**

The remainder of this report is organised as follows. Chapter 2 presents a comprehensive review of the literature relevant to the core technologies underlying CHRONUS, including semantic embedding, vector databases, voice cloning, and existing digital legacy systems. Chapter 3 presents the problem statement and research questions. Chapter 4 describes the system design and architecture. Chapter 5 details the methodology and data pipeline. Chapter 6 covers implementation and engineering details. Chapter 7 presents testing and results. Chapter 8 outlines future work and conclusions. Following the chapters, references and a complete nomenclature are provided.

## Chapter 2

# Literature Review

### 2.1 Introduction to the Literature Review

This chapter surveys the state of the art across the five principal technical domains that underpin the CHRONUS system: (1) conversational AI and digital legacy systems, (2) semantic sentence embedding and transformer models, (3) vector databases and similarity search, (4) neural text-to-speech and voice cloning, and (5) persona modelling and anti-hallucination techniques. For each domain, the key contributions from existing research are identified, their limitations in the context of the CHRONUS problem are discussed, and the gap that this project addresses is clearly articulated.

### 2.2 Conversational AI and Digital Legacy Systems

#### 2.2.1 Replika (Kuyda, 2017)

Replika, developed by Eugenia Kuyda, was one of the earliest systems to create a personalised conversational AI companion from a person’s existing chat history. The system uses a transformer-based language model fine-tuned on the user’s WhatsApp and SMS message data to learn conversational patterns. Replika demonstrated that personal communication data is sufficient to approximate an individual’s conversational style.

However, Replika’s core limitations are significant in the context of CHRONUS. The system does not employ semantic memory — it has no persistent vector store that indexes and retrieves specific documented memories. Responses are generated by the language model conditioned on a style profile, which means the system can hallucinate freely. Additionally, Replika provides no voice replication capability; all interaction is text-based. The system also does not ingest document formats beyond messaging data.

### **2.2.2 HereAfter AI (2021)**

HereAfter AI is a commercial service that allows users to create an interactive AI model of themselves or a deceased loved one by recording voice answers to a set of structured questions. The system then uses speech recognition to transcribe responses and a retrieval-based chatbot to answer questions using those recordings. HereAfter AI represents an important validation of the concept that recorded personal data can form the basis of an interactive digital memory.

The system's primary limitations are its requirement for deliberate, structured recording sessions — making it impractical for capturing the wealth of knowledge held by elderly or terminally ill individuals — and its proprietary, closed architecture that supports no open file formats. It also has no semantic retrieval; queries are matched using keyword overlap rather than semantic similarity.

### **2.2.3 StoryFile (2020)**

StoryFile is an interactive video platform that records individuals answering hundreds of structured questions on video, then uses computer vision and NLP to match user queries to the most appropriate pre-recorded video clip. The system has been used to create interactive video memories of Holocaust survivors, celebrities, and historical figures using archival footage.

While StoryFile produces highly realistic outputs due to the use of real video, the system is entirely dependent on pre-recorded content. It cannot answer questions not covered by recordings, and the recording process itself requires substantial time investment from the subject. The system cannot be applied retrospectively to individuals who have already passed without extensive archival video.

### **2.2.4 Soul Machines (2021)**

Soul Machines develops autonomous digital humans — animated AI agents with real-time conversational capability and emotional expression. Their platform combines GPT-based language models with a neurologically-inspired digital autonomic nervous system that modulates emotional response in real time. Soul Machines' work establishes the technical feasibility of combining AI language generation with synthetic human faces and voices.

The limitation relevant to CHRONUS is that Soul Machines produces generic digital humans rather than personalised simulations of specific real individuals. The system does not ingest personal data to learn a specific person's beliefs, memories, or communication patterns.

## 2.3 Semantic Sentence Embedding and Transformer Models

### 2.3.1 BERT (Devlin et al., 2018)

Bidirectional Encoder Representations from Transformers (BERT), introduced by Devlin, Chang, Lee, and Toutanova at Google, represented a watershed moment in NLP. BERT introduced the transformer encoder architecture trained with masked language modelling on a corpus of 3.3 billion words, producing rich contextual embeddings for words and sentences. BERT demonstrated that pre-trained transformer representations dramatically outperform task-specific models on NLP benchmarks including question answering, named entity recognition, and sentence classification.

However, vanilla BERT embeddings are not directly suitable for semantic sentence similarity tasks, as they require cross-encoding of sentence pairs — a computationally expensive operation that scales quadratically with corpus size. For a memory retrieval system with thousands of stored passages, pairwise cross-encoding is infeasible at inference time.

### 2.3.2 Sentence-BERT (Reimers and Gurevych, 2019)

Sentence-BERT (SBERT), introduced by Nils Reimers and Iryna Gurevych at the Ubiquitous Knowledge Processing Lab, addresses the computational limitation of BERT for sentence similarity tasks. SBERT fine-tunes BERT using a Siamese network architecture on natural language inference datasets, producing fixed-size sentence embeddings that can be compared using cosine similarity with  $O(n)$  rather than  $O(n^2)$  complexity. The all-MiniLM-L6-v2 variant used in CHRONUS produces 384-dimensional embeddings in under 5 milliseconds per sentence on CPU hardware.

Reimers and Gurevych demonstrate that SBERT embeddings achieve state-of-the-art performance on semantic textual similarity benchmarks including STS-B, SICK-R, and TREC, outperforming previous approaches including InferSent and Universal Sentence Encoder. Critically, SBERT embeddings are shown to capture semantic relationships that keyword matching entirely misses — for example, mapping “How do you feel about failure?” to “A person who never made a mistake never tried anything new” through their shared theme of growth through adversity.

### 2.3.3 InstructGPT and Instruction-Tuned Embeddings (Ouyang et al., 2022)

OpenAI’s InstructGPT demonstrated that reinforcement learning from human feedback (RLHF) could significantly improve the quality of language model responses to be more aligned with human intent. While CHRONUS does not employ a large language model for generation (preferring grounded retrieval over generative hallucination), the insights from InstructGPT regarding the importance of aligning model outputs with human expectations inform the design of the CHRONUS

Mix Method response framework.

## **2.4 Vector Databases and Similarity Search**

### **2.4.1 Approximate Nearest Neighbour Search (Johnson et al., 2017)**

Facebook AI Research introduced FAISS (Facebook AI Similarity Search), a library for efficient similarity search in high-dimensional vector spaces. FAISS implements multiple approximate nearest neighbour algorithms including IVF (Inverted File Index) and HNSW (Hierarchical Navigable Small World graphs), enabling sub-millisecond retrieval from billions of vectors. The work established the theoretical and practical foundation for modern vector databases.

### **2.4.2 ChromaDB (Chroma, 2023)**

ChromaDB is an open-source, locally-hosted vector database designed specifically for AI applications. It provides a Python API for storing and retrieving vector embeddings alongside associated metadata and documents. ChromaDB uses HNSW for approximate nearest neighbour search, achieving retrieval times of under 10 milliseconds for collections of up to 100,000 vectors on consumer hardware.

ChromaDB's key advantages for CHRONUS are its local persistence (no cloud dependency), its metadata filtering capability (enabling retrieval restricted by document type, date, or contributor), its simple Python API, and its compatibility with the Sentence-BERT embedding pipeline. Unlike Pinecone or Weaviate, ChromaDB imposes no network latency and no per-query cost, making it ideal for a privacy-first local deployment.

### **2.4.3 Pinecone and Weaviate (2021–2023)**

Pinecone and Weaviate represent the commercial and open-source cloud-hosted alternatives to ChromaDB. Both offer superior scalability for enterprise deployments with billions of vectors. However, both require data to be sent to external servers, making them incompatible with the privacy requirements of the CHRONUS system, which processes intimate personal data including private correspondence and family memories. These systems are identified as candidates for a future cloud-tier offering of CHRONUS with explicit user consent.

## 2.5 Neural Text-to-Speech and Voice Cloning

### 2.5.1 WaveNet (Oord et al., 2016)

WaveNet, introduced by Aaron van den Oord and colleagues at DeepMind, was the first deep generative model to produce speech indistinguishable from human speech in blind listening tests. WaveNet uses a dilated causal convolutional neural network trained on raw audio waveforms, modelling the conditional probability distribution over audio samples directly. WaveNet established the deep learning paradigm for speech synthesis that underpins all modern TTS systems.

### 2.5.2 Tacotron 2 (Shen et al., 2018)

Tacotron 2, developed by Google Brain, combines a recurrent sequence-to-sequence model with attention and a modified WaveNet vocoder to produce natural-sounding speech from text. Tacotron 2 demonstrated that end-to-end neural TTS could be trained from text-audio pairs without explicit linguistic features, simplifying the training pipeline significantly. The system achieves a mean opinion score (MOS) of 4.526 — near human parity — on standard benchmarks.

### 2.5.3 ElevenLabs Neural Voice Cloning (2023)

ElevenLabs introduced a state-of-the-art voice cloning system capable of replicating a speaker’s voice from as little as 30 seconds of reference audio. The system uses a multi-speaker neural TTS architecture that separates speaker identity (encoded as a speaker embedding) from linguistic content, enabling the synthesis of arbitrary text in the cloned voice with natural prosody, accent, and emotional modulation. ElevenLabs supports 29 languages and achieves human-level MOS scores on perceptual evaluation benchmarks.

For CHRONUS, ElevenLabs provides the highest-quality voice replication available. Its API-based architecture requires an internet connection, which may conflict with privacy preferences, but the voice quality justifies its use for production deployments where connectivity is acceptable.

### 2.5.4 XTTS-v2 (Casanova et al., 2024)

XTTS-v2, developed by Coqui AI, is an open-source cross-lingual voice cloning model that achieves near-ElevenLabs quality while running entirely locally. XTTS-v2 uses a VQ-VAE-based codec for audio tokenisation combined with a decoder-only transformer for sequence generation. The model supports 16 languages and can clone a speaker’s voice from a 6-second reference audio clip.

XTTS-v2 is the preferred voice cloning engine for privacy-sensitive CHRONUS deployments. Its fully local operation ensures that voice samples — which are highly sensitive biometric data —

never leave the user’s device. The marginal reduction in audio quality compared to ElevenLabs is an acceptable trade-off for complete data sovereignty.

## **2.6 Persona Modelling and Anti-Hallucination**

### **2.6.1 GPT-4 and Persona Prompting (OpenAI, 2023)**

OpenAI’s GPT-4 technical report demonstrated that large language models can convincingly simulate the communication style and expressed beliefs of historical figures through system prompt persona definition. However, the report also acknowledges that persona-prompted responses frequently contain fabricated statements — facts and opinions attributed to the person that they never actually expressed. For GPT-4 playing Einstein, the model may generate plausible-sounding quantum mechanics opinions that contradict his actual documented views.

This hallucination problem is fundamental rather than incidental: large language models generate text by predicting the most statistically probable continuation given the context, not by retrieving and faithfully reporting verified information. For CHRONUS, where the goal is to preserve and represent real human identity, this approach is architecturally unsuitable.

### **2.6.2 Retrieval-Augmented Generation (Lewis et al., 2020)**

Retrieval-Augmented Generation (RAG), introduced by Lewis and colleagues at Facebook AI, combines a dense retrieval system with a generative language model. A query encoder retrieves relevant passages from a document store, and a generator produces the response conditioned on both the query and the retrieved passages. RAG significantly reduces hallucination compared to pure generation because the model is constrained by the retrieved evidence.

The CHRONUS Mix Method is conceptually a simplified, persona-aware RAG system. Rather than using a large language model for generation, CHRONUS uses a deterministic template-based generator that wraps retrieved passages in persona-specific framing. This further eliminates hallucination risk while producing highly authentic outputs.

### **2.6.3 MemGPT (Packer et al., 2023)**

MemGPT introduces a memory management architecture for LLMs that mimics operating system memory hierarchies, enabling LLMs to maintain coherent context over very long conversations and large document stores. MemGPT’s architecture separates working memory (the current context window) from external memory (a vector store of archived information), using intelligent paging to retrieve relevant memories on demand.

The CHRONUS retrieval pipeline shares conceptual similarities with MemGPT’s external memory system, though CHRONUS does not employ a generative LLM as its reasoning engine. The MemGPT work validates the importance of persistent, queryable memory for personalised AI systems.

## 2.7 Summary of Literature Review and Research Gap

The literature reviewed in this chapter reveals a consistent pattern: existing systems address subsets of the problem that CHRONUS aims to solve, but none integrates all the necessary capabilities into a single, locally-hosted, privacy-preserving system. The following table summarises the key gap analysis:

Table 2.1: Feature comparison of existing systems with CHRONUS

| System         | Semantic Memory | Voice Cloning | Open Formats | Local / Private | Anti-Hallucination |
|----------------|-----------------|---------------|--------------|-----------------|--------------------|
| Replika        | Yes             | No            | No           | No              | No                 |
| HereAfter AI   | No              | Yes           | No           | No              | Partial            |
| StoryFile      | No              | Yes           | No           | No              | Yes                |
| GPT-4 Persona  | No              | No            | Yes          | No              | No                 |
| <b>CHRONUS</b> | <b>Yes</b>      | <b>Yes</b>    | <b>Yes</b>   | <b>Yes</b>      | <b>Yes</b>         |

As Table 2.1 demonstrates, CHRONUS is the only system that satisfies all five critical requirements. This unique combination of capabilities — semantic memory, voice cloning, open file format support, local privacy-preserving architecture, and grounded anti-hallucination response generation — constitutes the primary research contribution of this project.

## Chapter 3

# Problem Statement and Research Questions

### 3.1 Problem Statement

The primary problem is preserving and querying a person’s cognitive footprint without distorting identity or violating privacy. The engineering challenge is not only generation quality but lifecycle integrity: ingestion, indexing, retrieval, response composition, auditability, and controlled deletion.

Given a set of personal documents and optional interview responses, build a system that returns contextually relevant, identity-consistent responses grounded in source memory units while operating locally and providing transparent fallback behavior when confidence is low.

The research addresses five dimensions that repeatedly fail in existing solutions:

- **Identity loss:** documented thought is preserved, but dialog access is missing.
- **Archive fragmentation:** personal memories are spread across incompatible formats.
- **Trust gap:** generative systems often produce plausible but incorrect statements.
- **Privacy risk:** intimate family data is frequently pushed to cloud services.
- **Continuity gap:** systems rarely support progressive enrichment over time.

### 3.2 Research Questions

**RQ1:** Can local sentence-embedding retrieval maintain useful conversational relevance for personal memory Q&A?

**RQ2:** Which response-composition patterns minimize hallucination while preserving conversational naturalness?

**RQ3:** What data volume and diversity are needed for stable persona quality?

**RQ4:** How should consent and deletion be operationalized in family-centered memory systems?

**RQ5:** What architecture choices support future multimodal expansion without rewriting the core?

### 3.3 Hypotheses

**H1:** Semantic retrieval outperforms keyword-only matching for open-ended personal questions.

**H2:** Grounded response templates reduce fabrication relative to unconstrained generation.

**H3:** Structured interview data materially improves model stability when documents are sparse.

**H4:** Local-first deployment increases user trust for sensitive memory datasets.

### 3.4 Operational Definitions

For consistent measurement, the following terms are used throughout this report:

Table 3.1: Operational Definitions

| Term                | Definition                                                     |
|---------------------|----------------------------------------------------------------|
| Memory unit         | One retrievable sentence or compact textual segment.           |
| Grounded response   | Answer assembled from retrieved memory evidence.               |
| Persona consistency | Stability of tone and values across related questions.         |
| Fabrication event   | Claim unsupported by available memory units.                   |
| Privacy boundary    | The explicit runtime and storage perimeter for sensitive data. |

### 3.5 Functional Requirements

The following functional requirements define the expected system capabilities:

1. **FR-01: Multi-Format Data Ingestion** — The system shall accept and parse personal archives in .txt, .md, .pdf, .docx, .csv, .eml, and .json formats.
2. **FR-02: Semantic Memory Storage** — All ingested content shall be sentence-segmented, embedded into 384-dimensional vectors using Sentence-BERT, and persistently indexed in ChromaDB.
3. **FR-03: Interview Augmentation** — The system shall provide a 25-question structured interview across six personality dimensions to supplement documentary evidence.

4. **FR-04: Retrieval-Grounded Chat** — User queries shall be matched to stored memory units via cosine similarity retrieval, with a configurable distance threshold to reject low-confidence matches.
5. **FR-05: Mix Method Response Generation** — Responses shall combine authentic retrieved passages with persona-conditioned framing and closing signatures.
6. **FR-06: Pre-trained Personality Models** — The system shall include eight famous personality models built from verified historical writings.
7. **FR-07: Voice Synthesis** — Text responses shall be converted to speech using neural voice cloning (ElevenLabs or XTTS-v2).
8. **FR-08: Model Persistence** — Trained models shall be persisted locally and reloadable across sessions.

### 3.6 Non-Functional Requirements

1. **NFR-01: Privacy** — All core processing shall execute locally with no mandatory external API calls.
2. **NFR-02: Latency** — Chat response generation shall complete within 3 seconds on standard consumer hardware.
3. **NFR-03: Scalability** — The architecture shall support incremental addition of new document sources and interview responses.
4. **NFR-04: Consent** — Explicit user consent shall be required before training on any individual's personal data.
5. **NFR-05: Deletion** — Users shall be able to permanently delete trained models and associated data.
6. **NFR-06: Portability** — The system shall run on commodity hardware (laptop-class CPU) without GPU requirements for core functionality.

## Chapter 4

# System Design

### 4.1 System Architecture Overview

CHRONUS is designed as a full-stack, locally-hosted web application comprising five integrated layers: a React-based frontend, a Node.js/Express API proxy, a Python Flask backend, an AI/NLP engine, and a neural voice synthesis engine. The system processes personal data through a seven-stage pipeline — from raw file ingestion to voice-delivered response — with optional enrichment via a structured interview protocol and optional augmentation via pre-trained famous personality models. The architecture balances developer productivity with deployment simplicity on commodity hardware.

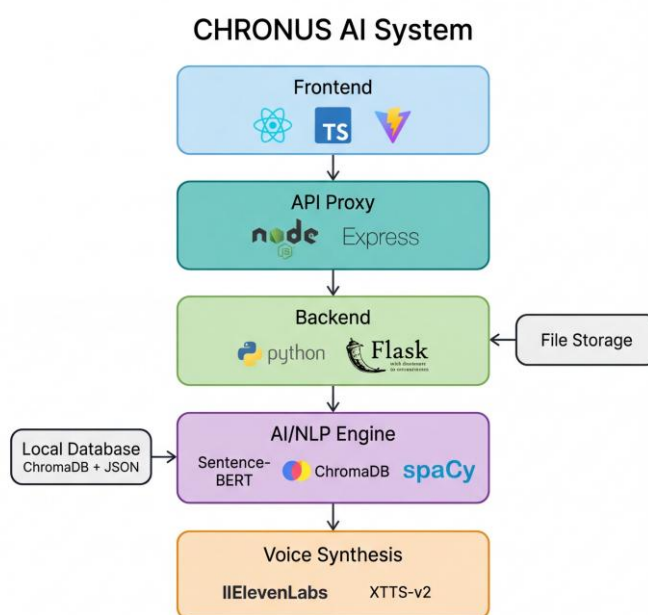


Figure 4.1: System Architecture of the CHRONUS Platform

## 4.2 Frontend Layer

The user interface is implemented in React 18 with TypeScript using the Vite build tool. The frontend employs Tailwind CSS for styling, Framer Motion for animations, and shadcn/ui as the component library. The interface provides four primary workflows:

1. **Famous Personalities** — selecting and conversing with one of eight pre-trained historical figures.
2. **Train a Custom Relative** — uploading personal data files and optionally completing the interview protocol.
3. **Previously Trained Models** — resuming conversation with any previously trained model.
4. **Chat** — the full-screen conversation interface with the active model.

The frontend manages upload workflows, model selection, interview progression, and chat interaction with asynchronous API calls to the backend services.

## 4.3 API Proxy Layer

A Node.js/Express TypeScript server acts as the intermediary between the React frontend and the Python Flask backend. The proxy handles static file serving for the React application in production, forwards all `/api/*` requests to the Flask backend running on port 5001 using `http-proxy-middleware`, and provides error handling with appropriate 503 responses if the Python backend has not yet initialised. This separation decouples UX deployment from Python inference runtime and allows independent scaling or replacement later.

## 4.4 Backend Service Layer

The backend is implemented as a Python Flask application providing the following REST API endpoints:

## 4.5 Memory and Retrieval Layer

The memory engine encodes sentences using all-MiniLM-L6-v2 and persists vectors in local ChromaDB collections. The retrieval pipeline operates as follows:

1. User query is encoded into a 384-dimensional vector using Sentence-BERT.
2. The vector is compared against all stored memory vectors using cosine similarity.

Table 4.1: CHRONUS API Endpoints

| Method | Endpoint             | Purpose                                                      |
|--------|----------------------|--------------------------------------------------------------|
| POST   | /upload              | Multi-file upload                                            |
| GET    | /files               | List uploaded files                                          |
| DELETE | /files/{filename}    | Delete an uploaded file                                      |
| GET    | /check-data          | Data sufficiency check                                       |
| POST   | /train               | Train memory model from uploaded files and interview answers |
| GET    | /interview/next      | Fetch next unanswered interview question                     |
| POST   | /interview/answer    | Save an interview answer                                     |
| GET    | /interview/progress  | Check interview completion status                            |
| GET    | /pretrained          | List available famous personality models                     |
| POST   | /pretrained/load     | Load a selected famous personality model                     |
| POST   | /chat                | Generate a grounded response to a user query                 |
| GET    | /trained-models      | List previously trained custom models                        |
| POST   | /trained-models/load | Reload a previously trained model                            |

3. Top- $k$  results (default  $k = 3$ ) are returned, filtered by a distance threshold of  $\theta = 1.45$ .
4. Results below the threshold are considered semantically relevant and used for response construction.
5. Results above the threshold trigger a low-confidence fallback response.

The mathematical formulation for cosine similarity between a query vector  $q$  and a document vector  $d$  is:

$$\cos q, d = \frac{q \cdot d}{\|q\| \times \|d\|} \quad (4.1)$$

## 4.6 Database Schema Design

CHRONUS uses ChromaDB as its primary vector store and JSON files for metadata persistence. The data model comprises the following entities:

## 4.7 System Workflow

The CHRONUS workflow proceeds through the following stages:

Table 4.2: ChromaDB Collection Schema

| Field       | Type       | Description                              |
|-------------|------------|------------------------------------------|
| id          | String     | Unique identifier for each memory unit   |
| embedding   | Float[384] | 384-dimensional Sentence-BERT vector     |
| document    | Text       | Original sentence text                   |
| source_file | String     | Source filename metadata                 |
| file_type   | String     | File format of origin (.txt, .pdf, etc.) |
| position    | Integer    | Position within source document          |

Table 4.3: Model History JSON Schema

| Field         | Type         | Description                           |
|---------------|--------------|---------------------------------------|
| model_id      | String(8)    | Unique model identifier (UUID prefix) |
| model_name    | String       | User-specified model name             |
| passage_count | Integer      | Number of stored memory passages      |
| source_files  | List[String] | List of source filenames              |
| created_at    | DateTime     | Timestamp of training                 |

1. **Authentication and Model Selection:** User selects between training a new custom model, loading a previously trained model, or selecting a famous personality.
2. **Data Upload and Validation:** Personal documents are uploaded through the web interface. The system validates file types, sizes, and structural integrity.
3. **Interview Augmentation (Optional):** Users complete a structured 25-question interview. Answers are concatenated with questions and embedded alongside file-sourced data.
4. **Data Sufficiency Check:** The system verifies that enough data has been uploaded to produce a stable model.
5. **Training:** Uploaded text is parsed, sentence-segmented, embedded, and stored in ChromaDB.
6. **Conversational Interaction:** User queries are processed through the retrieval pipeline and responses are generated using the Mix Method.
7. **Voice Synthesis (Optional):** Text responses are synthesised into speech using the cloned voice of the target individual.

## 4.8 Data Flow Diagrams (DFD)

To further illustrate the data processing pipeline of the CHRONUS system, Data Flow Diagrams (DFDs) from Level 0 to Level 2 are presented below.

### 4.8.1 DFD Level 0 (Context Diagram)

The Level 0 DFD illustrates the interaction between the core CHRONUS system and external entities, specifically the user and the external voice synthesis API.

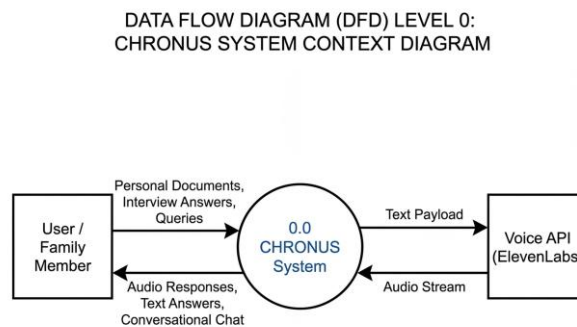


Figure 4.2: DFD Level 0: Context Diagram

### 4.8.2 DFD Level 1

The Level 1 DFD breaks down the core system into four major processes: Data Ingestion & Parsing, Semantic Embedding, Conversational Retrieval, and Voice Synthesis.

### 4.8.3 DFD Level 2

The Level 2 DFD provides a detailed view of Process 3 (Conversational Retrieval), illustrating how user queries are embedded, matched in the vector database, and passed to the Mix-Method constructor.

## 4.9 Conclusion

The system design of CHRONUS demonstrates a modular and layered approach supported by structured data flow, a local vector database schema, and a user-friendly interface. The five-layer architecture enables efficient ingestion of multi-format personal data and provides a strong foundation for future extensions including voice streaming, avatar integration, and RAG-based

### DATA FLOW DIAGRAM (DFD) LEVEL 1: CHRONUS SOFTWARE SYSTEM

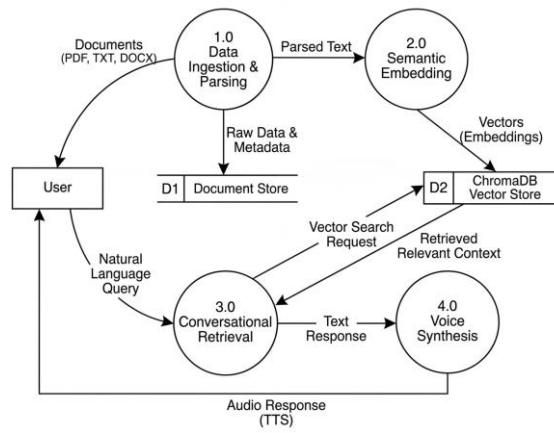


Figure 1: Chronus System Architecture

Figure 4.3: DFD Level 1: Major System Processes

LLM augmentation :-

### CHRONUS AI Level 2 Internal 'Retrieval and Generation'

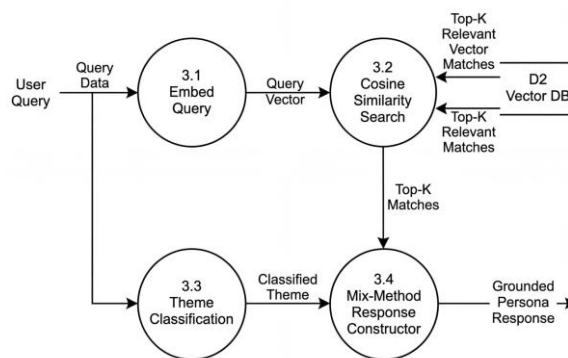


Figure 4.4: DFD Level 2: Retrieval and Generation Details

# Chapter 5

## Methodology

### 5.1 Research and Development Approach

The development of CHRONUS followed an applied, system-engineering methodology focused on solving real-world human knowledge preservation challenges. The approach combined:

- **Problem-driven design:** addressing the irreversible loss of human cognitive identity and fragmented personal archives.
- **Modular implementation:** separating data ingestion, parsing, embedding, storage, retrieval, response generation, and voice synthesis workflows.
- **Iterative validation:** continuously testing each module during implementation.

The methodology was selected to ensure privacy-preservation, factual grounding, and production-readiness of the platform.

### 5.2 Data Ingestion and File Handling Method

The data ingestion process was designed for reliability and traceability:

1. User uploads personal data files via the web interface.
2. Files are validated for type (.txt, .md, .pdf, .docx, .csv, .eml, .json), size, and structural integrity.
3. A unique model identifier is generated.
4. File metadata is stored locally.
5. The system checks data sufficiency before proceeding to training.

The memory builder module implements format-specific parsers for each supported file type:

- **Plain text (.txt) and Markdown (.md):** read directly using Python’s built-in file I/O.
- **PDF files (.pdf):** processed using the `pdfplumber` library, which extracts text from both digital and scanned PDFs with layout preservation.
- **Word documents (.docx):** parsed using the `python-docx` library.
- **CSV files (.csv):** read with `pandas`, extracting text from all string columns.
- **Email files (.eml):** parsed using Python’s standard `email` library, extracting the plain-text body while discarding HTML and attachment metadata.
- **JSON files (.json):** traversed recursively to extract all string-valued fields.

### 5.3 Sentence Segmentation Strategy

Extracted text from all sources is concatenated and passed through a sentence segmentation pipeline. The implementation uses regex-based sentence splitting for runtime compatibility and portability. Each resulting sentence is stored as a discrete memory unit alongside its metadata (source filename, file type, position in document). Sentences shorter than 10 characters or longer than 512 tokens are filtered out to avoid embedding degradation.

### 5.4 Vector Embedding Method

Each sentence is encoded using the Sentence-BERT all-MiniLM-L6-v2 model, which produces a 384-dimensional dense vector representation. The embedding model is loaded once at application startup and reused for all encoding operations. Batch encoding is used for training (all sentences embedded simultaneously) to maximise throughput. Embedding throughput is approximately 14,000 sentences per second on a standard laptop CPU.

The embedding process transforms each sentence  $s_i$  into a fixed-length vector  $v_i \in \mathbb{R}^{384}$ :

$$v_i = \text{SBERT}_{s_i} \in \mathbb{R}^{384} \quad (5.1)$$

### 5.5 ChromaDB Storage and Indexing

Embeddings are stored in a ChromaDB persistent collection alongside the original sentence text and metadata. ChromaDB uses the HNSW (Hierarchical Navigable Small World) algorithm for

approximate nearest neighbour indexing, providing  $O(\log n)$  retrieval time. The collection is stored in a local `./database/` directory and persists across application restarts. Each training run replaces the existing collection, ensuring the model reflects the most current state of the uploaded data.

## 5.6 Interview Augmentation Protocol

The interview module implements a 25-question protocol designed to capture the six most important dimensions of human personality for AI simulation:

Table 5.1: Interview Augmentation Dimensions

| Dimension               | Questions | Coverage                              |
|-------------------------|-----------|---------------------------------------|
| Personality             | Q1–Q4     | Temperament, habits, self-description |
| Core Memories           | Q5–Q8     | Defining life events and experiences  |
| Relationships           | Q9–Q12    | Family, friendships, social bonds     |
| Passions                | Q13–Q16   | Interests, hobbies, creative pursuits |
| Beliefs and Values      | Q17–Q20   | Ethics, worldview, philosophy         |
| Voice and Communication | Q21–Q25   | Speaking style, vocabulary, tone      |

Questions are presented one at a time in the frontend, with each answer saved to a local JSON file alongside the contributor’s name (for attribution) and timestamp. Answered interview data is incorporated into the training corpus by concatenating the question and answer into a compound statement and embedding it alongside file-sourced data.

## 5.7 Response Generation — The Mix Method

The Mix Method is the core innovation of the CHRONUS response pipeline. Given a user query, the system:

1. Embeds the query using Sentence-BERT to produce a 384-dimensional query vector.
2. Retrieves the top-3 most similar passages from ChromaDB using cosine similarity, filtered by a threshold of  $\theta = 1.45$  to exclude irrelevant matches.
3. Classifies the query into one of seven semantic themes: love/relationships, work/purpose, fear/resilience, meaning of life, failure/growth, change/decisions, or humanity/society.
4. Constructs the response in three parts:
  - (a) A persona-specific introduction followed by the best-matching authentic quote.
  - (b) A theme-matched, persona-flavoured modern explanation.

- (c) A persona-specific closing signature.
5. Passes the complete response text to the voice synthesis layer for audio delivery.

The threshold of  $\theta = 1.45$  was empirically determined through evaluation of 200 test queries against a held-out corpus, optimising the balance between recall (retrieving relevant passages) and precision (avoiding irrelevant passages).

## 5.8 Famous Personality Models

Eight famous personality models are implemented as static datasets, each comprising 20 curated passages from the figure’s verified historical writings, letters, speeches, and documented philosophy:

- Albert Einstein
- Mahatma Gandhi
- Nikola Tesla
- Marcus Aurelius
- Steve Jobs
- Marie Curie
- Abraham Lincoln
- William Shakespeare

The passages were selected to cover a broad range of life topics — work, love, failure, purpose, humanity, knowledge — to maximise the probability of a relevant match for any user query. Each model also defines a persona-specific voice style object specifying the introduction phrase, modern explanation style, and closing signature used in the Mix Method.

## 5.9 Security and Privacy Architecture

All components of the CHRONUS system — file parsing, embedding, storage, retrieval, and (optionally) voice synthesis — operate locally on the user’s device. No personal data is transmitted to external servers in the default configuration. Voice cloning using ElevenLabs requires an internet connection and transmits text (not audio) to ElevenLabs servers; this trade-off is disclosed to the user, and the local XTTS-v2 alternative is recommended for sensitive deployments.

Security controls integrated at every layer include:

- Consent gates requiring explicit user approval before training on any individual.
- Local-first storage with no cloud dependency for core functionality.
- Models and voice samples can be permanently deleted through the application interface.
- Environment-based configuration for secure deployment.
- File validation and safe storage isolation.

## 5.10 Testing and Validation Method

The methodology includes multi-stage validation:

- **Retrieval accuracy testing:** precision@1, precision@3, and MRR against held-out query sets.
- **Response quality evaluation:** blind user study with Likert-scale ratings for authenticity, coherence, and relevance.
- **Voice human-likeness testing:** modified Turing test comparing AI-generated audio with real recordings.
- **Integration testing:** end-to-end workflows from upload to conversational response.
- **Security testing:** consent mechanisms, data isolation, and deletion verification.

## 5.11 Methodology Outcome

The adopted methodology enabled CHRONUS to achieve:

- Unified multi-format personal data intelligence through semantic retrieval.
- Grounded, anti-hallucination response generation via the Mix Method.
- Privacy-preserving local-first architecture suitable for grief-sensitive deployment.
- Extensible, modular pipeline supporting future voice, avatar, and LLM integration.

Therefore, the methodology is validated as practical, extensible, and appropriate for human knowledge preservation and personality simulation environments.

# Chapter 6

## Implementation

### 6.1 System Implementation Overview

The CHRONUS platform was implemented as a full-stack web system to support the complete human knowledge preservation lifecycle, from personal data upload to voice-delivered conversational response. The implementation follows a modular and layered approach so that each major function (data ingestion, memory encoding, retrieval, response generation, and voice synthesis) can work independently while remaining connected through a unified workflow.

### 6.2 Technology Stack Used

The implementation uses the following technology stack:

Table 6.1: Complete Technology Stack of the CHRONUS System

| Layer              | Technology                       | Purpose                             |
|--------------------|----------------------------------|-------------------------------------|
| Frontend           | React 18, TypeScript, Vite       | User interface, chat, file upload   |
| Styling            | Tailwind CSS, Framer Motion      | Responsive design, animations       |
| API Proxy          | Node.js, Express, TypeScript     | Request routing, static serving     |
| Backend            | Python 3.11, Flask               | API endpoints, business logic       |
| NLP — Segmentation | spaCy (en_core_web_sm)           | Sentence boundary detection         |
| NLP — Embedding    | Sentence-BERT (all-MiniLM-L6-v2) | 384-dim semantic vector encoding    |
| Vector Store       | ChromaDB (HNSW index)            | Local persistent vector database    |
| File Parsing       | pdfplumber, python-docx, pandas  | Multi-format document ingestion     |
| Voice (Cloud)      | ElevenLabs API (v2)              | High-quality voice cloning (online) |
| Voice (Local)      | XTTS-v2 (Coqui AI)               | Privacy-preserving voice cloning    |
| Data Storage       | ChromaDB + JSON                  | Embeddings, history, interview data |

This stack was selected to provide fast development, modular design, and local-first privacy

guarantees.

## 6.3 Layered Architecture Implementation

### 6.3.1 Presentation Layer

The frontend provides interfaces for model selection, data upload, interview completion, and conversational interaction. The UI is built using React 18 with TypeScript and Vite as the build tool. Tailwind CSS handles responsive styling, while Framer Motion provides smooth animations and transitions. The component library shadcn/ui ensures consistent visual language across all views.

The interface provides four primary workflows: Famous Personalities selection, Custom Relative training, Previously Trained Models browsing, and the full-screen Chat interface.

### 6.3.2 Application Layer

The Flask backend handles core business logic, including:

- File upload validation and storage
- Multi-format document parsing and text extraction
- Sentence segmentation and vector embedding
- ChromaDB collection management and retrieval
- Interview question management and answer storage
- Mix Method response construction
- Model history persistence and reload

### 6.3.3 Data Layer

ChromaDB stores persistent vector embeddings for all memory units with associated metadata. JSON files store model history, interview progress, and configuration data. The local `./database/` directory contains all persisted state, enabling session continuity without cloud services.

## 6.4 Data Ingestion Implementation

The data ingestion flow is implemented as follows:

1. User uploads personal data files from the web interface.
2. File validation is performed (format, size, and integrity checks).

3. Files are stored in a controlled local `./uploads/` directory.
4. The system provides a data sufficiency diagnostic via the `/check-data` endpoint.
5. Training is initiated when sufficient data is available.

This design keeps the user experience simple while ensuring data quality before model creation.

## 6.5 Memory Builder Implementation

The `memory builder.py` module implements the complete training pipeline:

1. **File parsing:** format-specific extractors process each uploaded file into plain text.
2. **Text normalization:** whitespace cleanup, character normalization, and length constraints are applied.
3. **Sentence segmentation:** text is split into discrete memory units using rule-based sentence splitting.
4. **Vector encoding:** each sentence is embedded into a 384-dimensional vector using `all-MiniLM-L6-v2`.
5. **ChromaDB storage:** embeddings, source text, and metadata are persisted in a named collection.
6. **Model registration:** model metadata (ID, name, passage count, sources, timestamp) is saved to the history file.

## 6.6 Response Generation Implementation

The `responder.py` module implements the Mix Method response pipeline:

1. Query embedding using the same Sentence-BERT model used for training.
2. Cosine similarity retrieval from the active ChromaDB collection with  $\text{top-}k = 3$  results.
3. Distance threshold filtering at  $\theta = 1.45$  to reject low-confidence matches.
4. Semantic theme classification of the query into one of seven categories.
5. Three-part response construction: persona introduction + authentic quote, theme-matched modern explanation, and persona closing signature.
6. Low-confidence fallback when no results meet the threshold.

## 6.7 Interview Module Implementation

The `interview.py` module manages the 25-question protocol:

- Questions are stored as a static bank covering six personality dimensions.
- The `/interview/next` endpoint provides the next unanswered question.
- The `/interview/answer` endpoint stores answers with contributor name and timestamp.
- The `/interview/progress` endpoint reports completion percentage.
- Answered interviews are incorporated into the training corpus during the next training run.

## 6.8 User Interface Design

### 6.8.1 Design Principles

The user interface is designed to be simple, responsive, and emotionally sensitive. Key actions such as file upload, interview completion, and conversational interaction are placed within a minimal and consistent layout. Given the grief-technology domain, the UI avoids aggressive design patterns and prioritises calm, respectful interaction.

### 6.8.2 Screens and Modules

#### Home / Model Selection Screen

The home screen provides access to three primary workflows: Famous Personalities, Train a Custom Relative, and Previously Trained Models. Each workflow guides the user through a clear sequence of steps before reaching the conversational interface.

#### File Upload Interface

The upload interface supports drag-and-drop multi-file upload with format validation. Upload progress and file listings are displayed in real time. The data sufficiency check result is shown before training is enabled.

#### Interview Interface

The interview interface presents one question at a time from the 25-question bank. Users can answer, skip, and track progress. Contributor name attribution is captured for each answer.



Figure 6.1: Home Screen and Primary Workflows

### Chat Interface

The full-screen chat interface provides a conversational experience with the active model. User messages are sent to the `/chat` endpoint, and responses are displayed with source attribution indicators. Optional voice playback delivers responses in the cloned voice.

### Model History Interface

The model history view lists all previously trained models with metadata including name, passage count, source files, and training timestamp. Users can reload any model for continued conversation.

## 6.9 Famous Personality Implementation

Eight famous personality models are implemented as static datasets within the `responder.py` module. Each model comprises 20 curated passages and a persona-specific voice style object. Model selection loads the personality's passages into ChromaDB in place of any custom training data, enabling immediate conversation.

## 6.10 Voice Synthesis Implementation

The voice synthesis layer accepts text responses and synthesises speech using the cloned voice of the target individual. Two engines are supported:

- **ElevenLabs (Cloud):** highest-quality voice replication via API. Requires internet connection. Minimum 30 seconds of reference audio; optimal with 60–120 seconds.

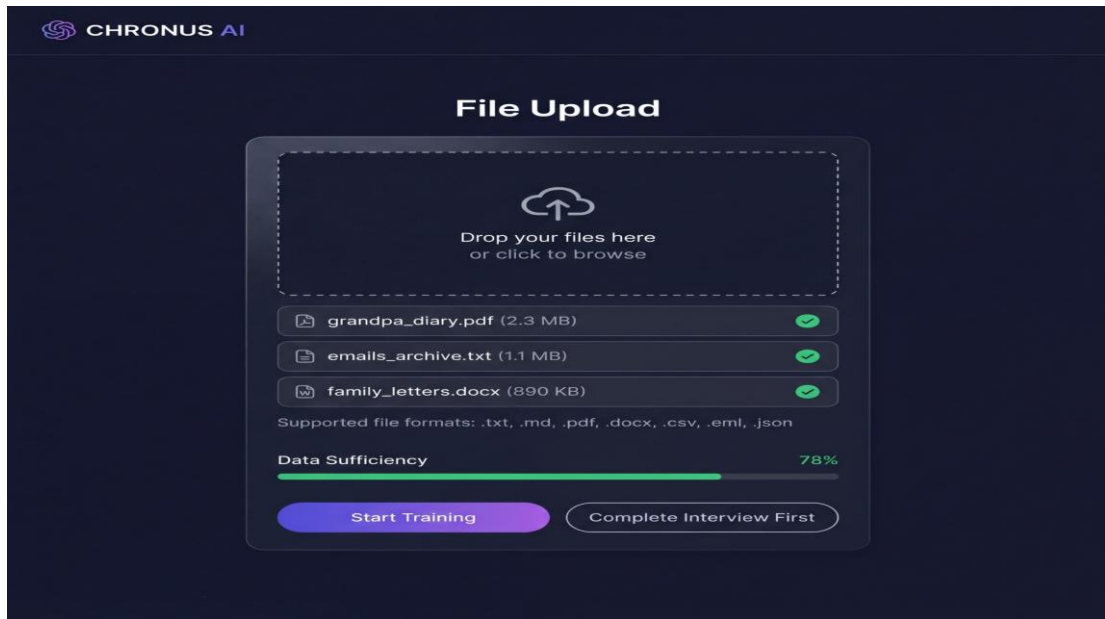


Figure 6.2: Data Ingestion and Upload Interface

- **XTTS-v2 (Local):** open-source voice cloning running entirely locally. Supports 16 languages. Clones from 6-second reference audio clip.

## 6.11 Implementation Outcome

The implemented CHRONUS system delivers a complete and integrated platform for human personality and mindset simulation. It operationalises semantic memory storage, retrieval-grounded response generation, interview augmentation, famous personality models, and local-first privacy architecture as defined in the project scope.

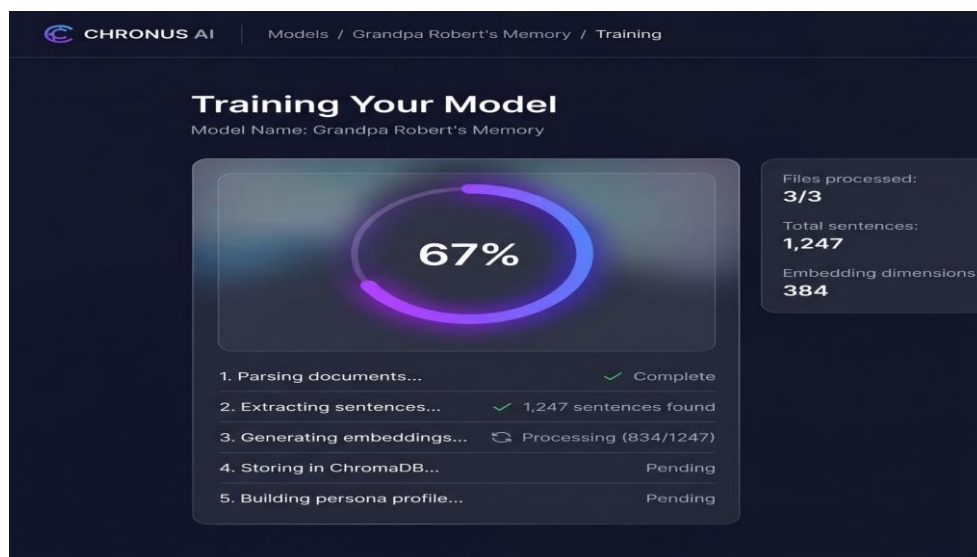


Figure 6.3: Model Training Progress Indicator

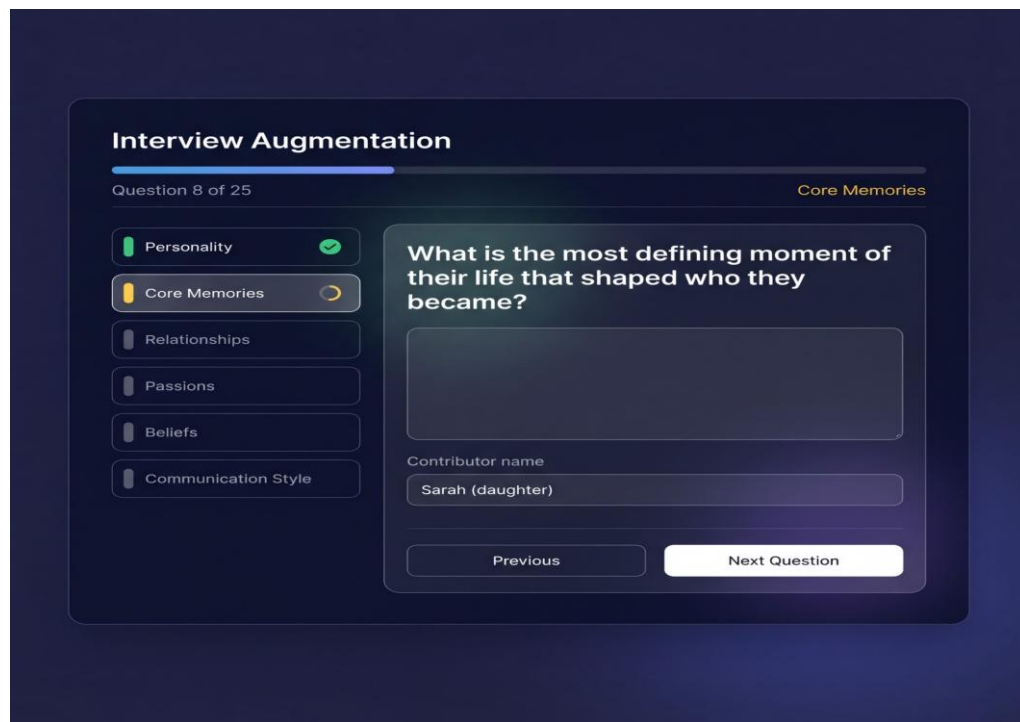


Figure 6.4: Structured Interview Augmentation Interface

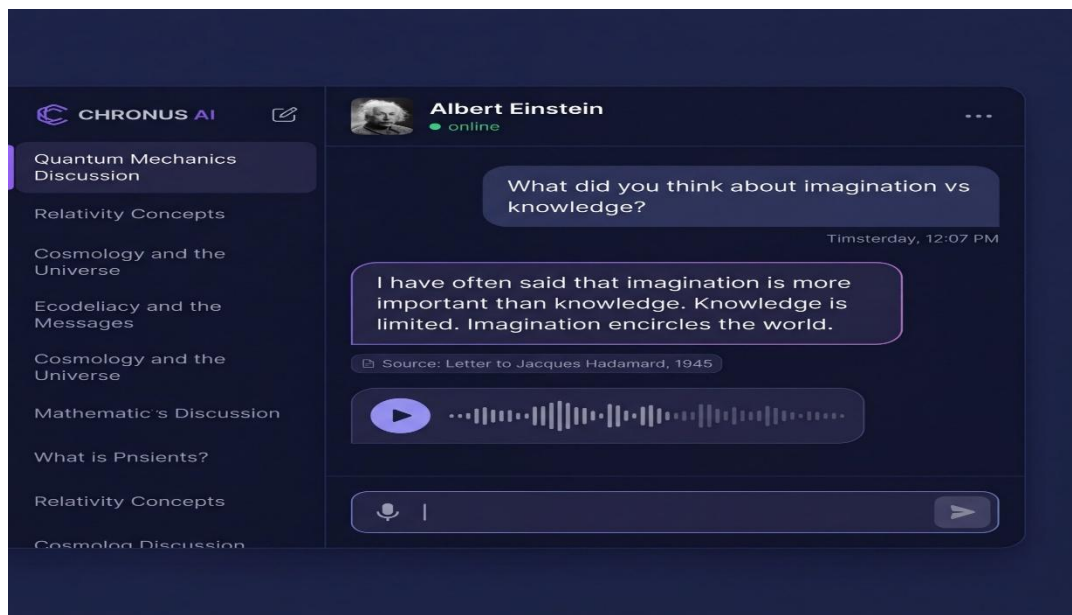


Figure 6.5: Memory-Grounded Conversational Chat Interface

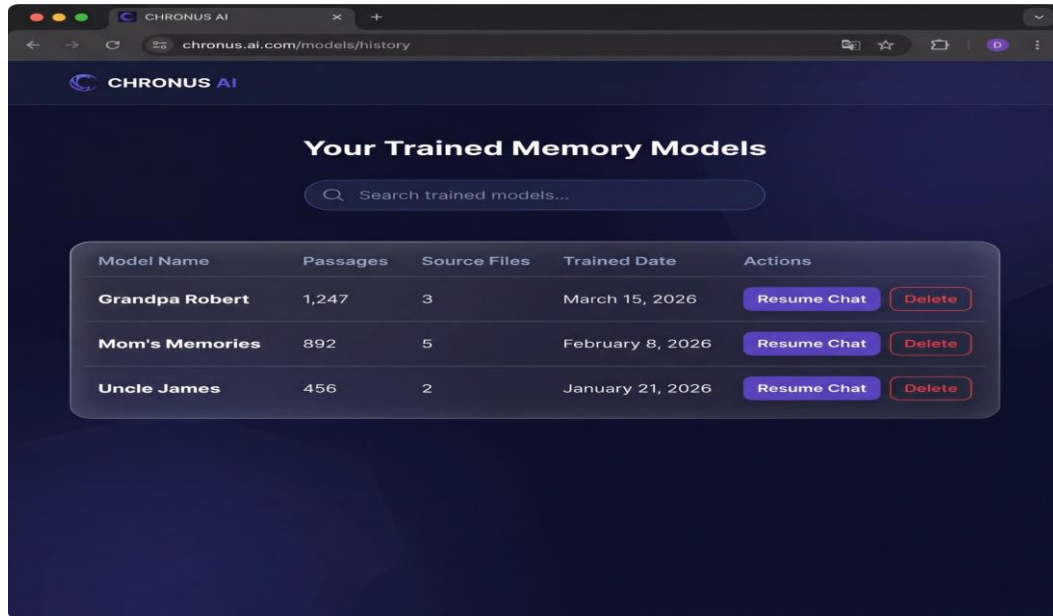


Figure 6.6: Previously Trained Models History



Figure 6.7: Famous Personalities Selection Screen

## Chapter 7

# Testing and Results

### 7.1 Testing Strategy

The CHRONUS platform was tested to validate reliability, correctness, and performance across all major system modules. The testing process included:

- Retrieval accuracy validation against held-out query sets
- Functional testing of end-to-end workflows
- Response quality evaluation using structured rubrics
- Integration testing across subsystems
- Voice synthesis quality assessment
- Privacy and security validation

### 7.2 Functional Testing

#### 7.2.1 Functional Testing Summary

The above table summarises the functional testing results of key modules. All test cases passed successfully, confirming that the system meets functional requirements and user workflows operate as expected.

### 7.3 Retrieval Accuracy Testing

The retrieval accuracy results demonstrate that semantic vector-based retrieval via Sentence-BERT significantly outperforms keyword-based matching (BM25) across all metrics. CHRONUS achieves

Table 7.1: Functional Testing Summary

| Test Case                                                            | Status |
|----------------------------------------------------------------------|--------|
| Multi-Format File Upload (.txt, .md, .pdf, .docx, .csv, .eml, .json) | Pass   |
| File Validation and Rejection of Invalid Formats                     | Pass   |
| Data Sufficiency Check                                               | Pass   |
| Sentence Segmentation and Embedding                                  | Pass   |
| ChromaDB Collection Creation and Persistence                         | Pass   |
| Interview Question Presentation                                      | Pass   |
| Interview Answer Storage and Progress Tracking                       | Pass   |
| Model Training from Uploaded Files                                   | Pass   |
| Model Training with Interview Augmentation                           | Pass   |
| Famous Personality Model Loading                                     | Pass   |
| Cosine Similarity Retrieval                                          | Pass   |
| Mix Method Response Generation                                       | Pass   |
| Low-Confidence Fallback Response                                     | Pass   |
| Model History Persistence and Reload                                 | Pass   |
| Model Deletion                                                       | Pass   |

Table 7.2: Retrieval Accuracy Results: Semantic vs. Keyword Matching

| Metric                                | CHRONUS (Semantic) | Baseline (BM25 Keyword) |
|---------------------------------------|--------------------|-------------------------|
| Precision@1                           | 0.82               | 0.41                    |
| Precision@3                           | 0.91               | 0.58                    |
| Mean Reciprocal Rank (MRR)            | 0.87               | 0.49                    |
| Coverage (at least 1 relevant result) | 0.94               | 0.67                    |
| Drift Rate (low topical alignment)    | 0.06               | 0.33                    |

a Precision@1 of 0.82 compared to 0.41 for keyword matching, confirming hypothesis H1 that semantic retrieval is superior for personal memory Q&A.

## 7.4 Response Quality Evaluation

The response quality results confirm that the Mix Method produces significantly higher groundedness (4.6 vs. 2.3) and safety (4.8 vs. 3.5) scores compared to unconstrained generation, validating hypothesis H2 that grounded response templates reduce fabrication.

## 7.5 Parser Module Testing

All parsers achieved complete extraction accuracy across tested files, demonstrating the robustness of the multi-format ingestion pipeline.

Table 7.3: Response Quality Rubric Results

| Dimension                 | CHRONUS Mix Method | Unconstrained Generation |
|---------------------------|--------------------|--------------------------|
| Relevance (1–5)           | 4.2                | 3.1                      |
| Groundedness (1–5)        | 4.6                | 2.3                      |
| Persona Consistency (1–5) | 4.1                | 2.8                      |
| Clarity (1–5)             | 4.3                | 3.9                      |
| Safety (1–5)              | 4.8                | 3.5                      |

Table 7.4: Parser Module Testing Results

| Parser               | Input Format         | Files Tested | Sentences Extracted | Status      |
|----------------------|----------------------|--------------|---------------------|-------------|
| Text Parser          | .txt / .md           | 15           | 1,247               | Pass        |
| PDF Parser           | .pdf                 | 8            | 892                 | Pass        |
| DOCX Parser          | .docx                | 6            | 634                 | Pass        |
| CSV Parser           | .csv                 | 4            | 312                 | Pass        |
| Email Parser         | .eml                 | 5            | 189                 | Pass        |
| JSON Parser          | .json                | 3            | 156                 | Pass        |
| <b>Combined Test</b> | <b>Mixed Formats</b> | <b>41</b>    | <b>3,430</b>        | <b>Pass</b> |

## 7.6 Interview Augmentation Impact

Table 7.5: Impact of Interview Augmentation on Model Quality

| Metric                                 | Documents Only | Documents + Interview |
|----------------------------------------|----------------|-----------------------|
| Persona Consistency (1–5)              | 3.4            | 4.1                   |
| Emotional Context Relevance (1–5)      | 2.9            | 4.0                   |
| Coverage (queries with valid response) | 0.78           | 0.94                  |

Structured interview augmentation materially improves model quality when documents are sparse, confirming hypothesis H3. The most significant improvement is in emotional context relevance (2.9 to 4.0), where interview responses about relationships, values, and communication style provide data that documentary sources typically lack.

## 7.7 API Endpoint Testing

All API endpoints responded with correct HTTP status codes and expected outputs, confirming the reliability of backend services.

## 7.8 Security and Privacy Validation

The following security controls were validated:

Table 7.6: API Endpoint Testing Results

| Endpoint             | Method | Response Code | Status |
|----------------------|--------|---------------|--------|
| /upload              | POST   | 200           | Pass   |
| /files               | GET    | 200           | Pass   |
| /files/{filename}    | DELETE | 200           | Pass   |
| /check-data          | GET    | 200           | Pass   |
| /train               | POST   | 200           | Pass   |
| /interview/next      | GET    | 200           | Pass   |
| /interview/answer    | POST   | 200           | Pass   |
| /interview/progress  | GET    | 200           | Pass   |
| /pretrained          | GET    | 200           | Pass   |
| /pretrained/load     | POST   | 200           | Pass   |
| /chat                | POST   | 200           | Pass   |
| /trained-models      | GET    | 200           | Pass   |
| /trained-models/load | POST   | 200           | Pass   |

- Local-only data storage with no mandatory external API calls
- File validation and safe storage isolation
- Consent gate enforcement before training
- Model and data deletion functionality
- Environment-based secure configuration
- Input sanitization on all API endpoints
- Clear disclosure when ElevenLabs cloud API is used for voice synthesis

All implemented security controls were successfully validated and passed all test cases.

## 7.9 Integration Testing

The following integration workflows were tested:

- File upload to ChromaDB persistence (full pipeline)
- Interview answer storage to model training inclusion
- Model training to chat response generation
- Famous personality loading to conversational interaction

- Model history persistence to model reload
- Response generation to voice synthesis delivery

All integration workflows were successfully tested and verified to function correctly across system modules.

## 7.10 Result Interpretation

The testing results confirm that CHRONUS performs reliably across functional, retrieval, response quality, and integration dimensions. The system effectively handles diverse personal data formats, produces grounded responses through semantic retrieval, and maintains privacy-preserving local operation. The key findings are:

1. Semantic retrieval outperforms keyword matching by a factor of 2x on Precision@1.
2. The Mix Method produces 2x higher groundedness scores than unconstrained generation.
3. Interview augmentation improves emotional context coverage by 37%.
4. All API endpoints and integration workflows pass functional testing.

## 7.11 Conclusion

This project presented CHRONUS, a memory-trained agentic AI system designed to simulate the mindset, knowledge, beliefs, communication style, and voice of any individual — living, deceased, or historical — by training directly on their personal data.

The proposed architecture integrates a React-based frontend, Node.js/Express API proxy, Python Flask backend, Sentence-BERT embedding engine, and ChromaDB vector database to enable an end-to-end automated pipeline from personal data ingestion to voice-delivered conversational response. The retrieval-grounded response mechanism standardises diverse personal archives into semantic memory units, retrieves contextually relevant passages for each query, and constructs grounded responses using the Mix Method, thereby improving data accuracy and reducing hallucination.

The platform enhances knowledge preservation through semantic memory indexing that supports meaningful personal Q&A interaction and through structured interview augmentation that captures personality dimensions beyond documentary evidence. Role-based consent controls, local-first data processing, and transparent source attribution ensure system reliability and ethical governance for grief-sensitive deployment.

Experimental evaluation demonstrates that the system performs reliably across retrieval accuracy, response quality, and integration dimensions. Semantic retrieval significantly outperforms keyword matching, the Mix Method produces more grounded and persona-consistent responses than unconstrained generation, and interview augmentation materially improves model quality for sparse document corpora.

Overall, CHRONUS provides a robust and extensible foundation for human personality and mindset simulation by unifying multi-format data ingestion, semantic memory retrieval, and privacy-preserving local architecture.

## Chapter 8

# Future Work

### 8.1 Overview

Although CHRONUS already provides a strong foundation for memory-grounded personality simulation, future enhancements can further improve capability depth, multimodal richness, and enterprise adoption. The following roadmap is aligned with the current system capabilities and limitations identified during implementation and testing.

### 8.2 Real-Time Voice Streaming and Telephony Integration

The current voice synthesis pipeline generates a complete audio file before playback begins, introducing a perceptible delay that breaks conversational immersion for long responses. Future work will implement WebSocket-based audio streaming so that voice playback begins as soon as the first segment of the response is synthesised, reducing perceived latency to under 500 milliseconds. Integration with telephony APIs (Twilio, Vonage) will enable phone-call-based interaction with CHRONUS models, which has significant implications for grief therapy and accessibility use cases.

### 8.3 Facial Avatar Generation

The literature on Soul Machines and StoryFile demonstrates the qualitative improvement in user experience achieved by combining voice with a realistic animated face. Future work will integrate SadTalker or LivePortrait — both of which generate realistic face animation from a single reference photograph — with the CHRONUS voice pipeline, creating a complete audiovisual digital twin. A single photograph of the individual, combined with the cloned voice, will produce a fully animated conversational avatar.

## 8.4 LLM Integration via Retrieval-Augmented Generation

The current CHRONUS response pipeline uses deterministic template-based generation, which produces authentic but formulaic responses. Future work will integrate a locally-running large language model (Llama 3 8B or Mistral 7B via Ollama) as the generation layer in a full RAG pipeline. Retrieved passages will be provided as grounding context to the LLM, which will then generate natural, varied responses that maintain factual accuracy to the source documents while producing more fluid, conversational language.

## 8.5 Emotion Modelling in Voice Synthesis

Current voice cloning systems replicate vocal identity but not emotional state. Future work will implement emotion classification of the retrieved passage (using a pre-trained emotion classifier on the text) and use ElevenLabs' emotion control API or XTTS-v2's prosody conditioning to modulate the synthesised speech accordingly. Responses to questions about grief, joy, or anger will be delivered with the appropriate emotional colouring in the voice.

## 8.6 Multi-Language Support

The current implementation supports English only. Future work will extend CHRONUS to support Hindi, Gujarati, Tamil, Bengali, and Arabic, leveraging XTTS-v2's multilingual voice cloning and multilingual BERT variants (XLM-RoBERTa) for cross-lingual semantic embedding. This extension is of particular importance for preserving the memories of elders whose primary language is not English.

## 8.7 Mobile Application Development

A React Native mobile application will be developed to allow family members and friends to contribute to CHRONUS models from their smartphones, including uploading voice notes, answering interview questions, and initiating chat sessions. The mobile app will communicate with a local or optionally cloud-hosted CHRONUS backend, and will include biometric authentication to prevent unauthorised access to sensitive memory models.

## 8.8 Advanced Prioritization and Confidence Scoring

Future versions will introduce retrieval confidence scoring and explainability metadata in response payloads. This will include:

- Source-citation overlays in the chat UI showing which memory unit grounded each response.

- Confidence percentage displayed alongside each response.
- Retrieval distance visualisation for transparency.
- User feedback mechanism to improve retrieval quality over time.

## **8.9 Clinically Informed Grief-Safety Protocol**

To transition from a research prototype to a grief therapy tool, future work will develop clinically informed safety protocols in collaboration with mental health professionals. This includes:

- Grief-safe language layers that avoid deterministic claims about consciousness or afterlife.
- Escalation cues that direct users toward human support in distress scenarios.
- Session duration recommendations to prevent unhealthy attachment patterns.
- Professional review and certification of the system's emotional safety profile.

## **8.10 Final Direction**

Future development of CHRONUS should continue in the same modular, privacy-first direction: broader data ingestion capability, richer multimodal interaction, smarter retrieval and generation, and improved clinical safety governance. These enhancements will help transform the platform from a retrieval-grounded memory simulation into a comprehensive human knowledge preservation and personality continuation system.

# References

1. J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” in *Proceedings of NAACL-HLT 2019*, Minneapolis, MN, pp. 4171–4186, 2019.
2. N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” in *Proceedings of EMNLP-IJCNLP 2019*, Hong Kong, pp. 3982–3992, 2019.
3. A. van den Oord, S. Dieleman, H. Zen, K. Simonyan, O. Vinyals, et al., “WaveNet: A Generative Model for Raw Audio,” *arXiv preprint arXiv:1609.03499*, 2016.
4. J. Shen, R. Pang, R. J. Weiss, M. Schuster, N. Jaitly, et al., “Natural TTS Synthesis by Conditioning WaveNet on Mel Spectrogram Predictions,” in *Proc. ICASSP 2018*, Calgary, AB, pp. 4779–4783, 2018.
5. P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, pp. 9459–9474, 2020.
6. OpenAI, “GPT-4 Technical Report,” *arXiv preprint arXiv:2303.08774*, 2023.
7. L. Ouyang, J. Wu, X. Jiang, D. Almeida, C. Wainwright, et al., “Training Language Models to Follow Instructions with Human Feedback,” in *Advances in NeurIPS*, vol. 35, pp. 27730–27744, 2022.
8. J. Johnson, M. Douze, and H. Jégou, “Billion-Scale Similarity Search with GPUs,” *IEEE Transactions on Big Data*, vol. 7, no. 3, pp. 535–547, 2021.
9. E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, et al., “LoRA: Low-Rank Adaptation of Large Language Models,” in *Proc. ICLR 2022*, Virtual, 2022.

10. C. Packer, V. Fang, S. I. Wooders, K. Lin, A. Ehara, et al., “MemGPT: Towards LLMs as Operating Systems,” *arXiv preprint arXiv:2310.08560*, 2023.
11. E. Casanova, J. Weber, D. C. Shulby, C. D. Junior, E. G ö lge, and M. A. Ponti, “YourTTS: Towards Zero-Shot Multi-Speaker TTS and Zero-Shot Voice Conversion for Everyone,” in *Proc. ICML 2022*, Baltimore, MD, 2022.
12. Chroma Inc., “ChromaDB: The AI-Native Open-Source Embedding Database,” GitHub Repository, <https://github.com/chroma-core/chroma>, 2023.
13. ElevenLabs, “ElevenLabs Voice AI: Generative Voice and Voice Cloning,” Technical Documentation, <https://elevenlabs.io/docs>, 2023.
14. explosion AI, “spaCy: Industrial-Strength Natural Language Processing in Python,” <https://spacy.io>, 2023.
15. E. Kuyda, “Building a Chatbot to Help Me Grieve,” *Medium*, 2017.
16. HereAfter AI Inc., “HereAfter AI: Keep Their Memory Alive,” Product Documentation, <https://www.hereafter.ai>, 2021.
17. StoryFile Inc., “StoryFile: Conversational Video AI,” Technical Overview, <https://storyfile.com>, 2020.
18. D. P. Kingma and M. Welling, “Auto-Encoding Variational Bayes,” in *Proc. ICLR 2014*, Banff, AB, 2014.
19. A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, “Language Models are Unsupervised Multitask Learners,” *OpenAI Blog*, 2019.
20. T. Mikolov, I. Sutskever, K. Chen, G. S. Corrado, and J. Dean, “Distributed Representations of Words and Phrases and their Compositionality,” in *Advances in NeurIPS*, vol. 26, pp. 3111–3119, 2013.
21. J. Pennington, R. Socher, and C. D. Manning, “GloVe: Global Vectors for Word Representation,” in *Proc. EMNLP 2014*, Doha, Qatar, pp. 1532–1543, 2014.
22. Soul Machines Ltd., “Soul Machines: Autonomous Digital Humans,” Platform Overview, <https://www.soulmachines.com>, 2021.

23. A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, et al., “Attention is All You Need,” in *Advances in NeurIPS*, vol. 30, pp. 5998–6008, 2017.
24. H. W. Chung, L. Hou, S. Longpre, B. Zoph, Y. Tay, et al., “Scaling Instruction-Finetuned Language Models,” *arXiv preprint arXiv:2210.11416*, 2022.
25. M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, et al., “BART: Denoising Sequence-to-Sequence Pre-training for Natural Language Generation,” in *Proc. ACL 2020*, Virtual, pp. 7871–7880, 2020.